

Inrush Current

Lab 18 | ECEN 3730 - Henry Matar

Lab Overview

- Using a 2.2Ω sense resistor to measure the total current through the entire circuit
- Using the Math Function on the O-Scope calculating the difference between $V_{\text{sen}} +/ -$
- Utilize a 555 timer to draw current
- Measure the switching current

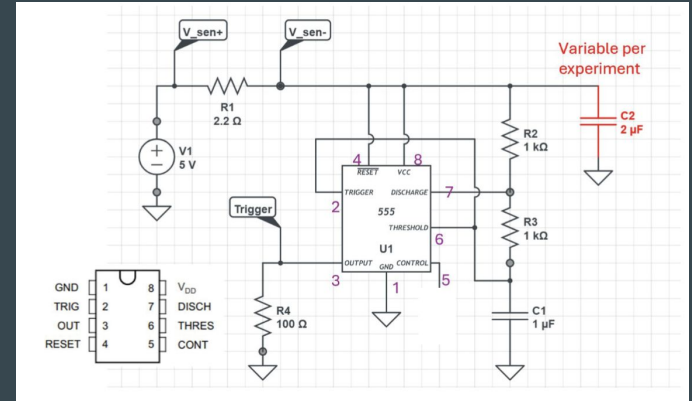


Fig. 1 Circuit Schematic Setup

Measured Results - Steady State Averaging



Fig. 2 Normal



Fig. 3 Averaging N=6

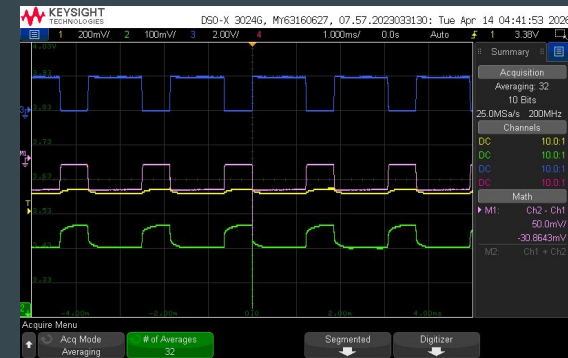


Fig. 4 Averaging N=32

Using the output of the 555 timer to trigger the scope the voltage is hard to read using the math function. Therefore, I use the averaging function to improve readability. Here we can measure the current through the resistor (2.2Ω). Using the $N=32$ Averaging one can see the measured current is 30.82mV, giving a current of 14mA.

Measured Results - Inrush Transient

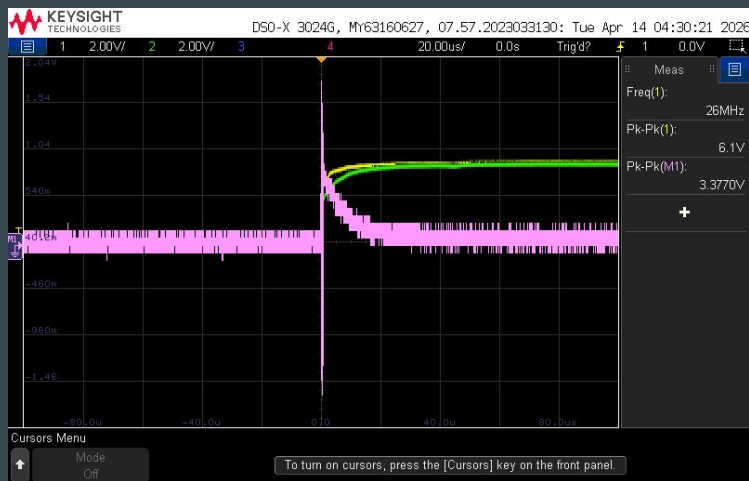


Fig. 5 1uF Decoupling

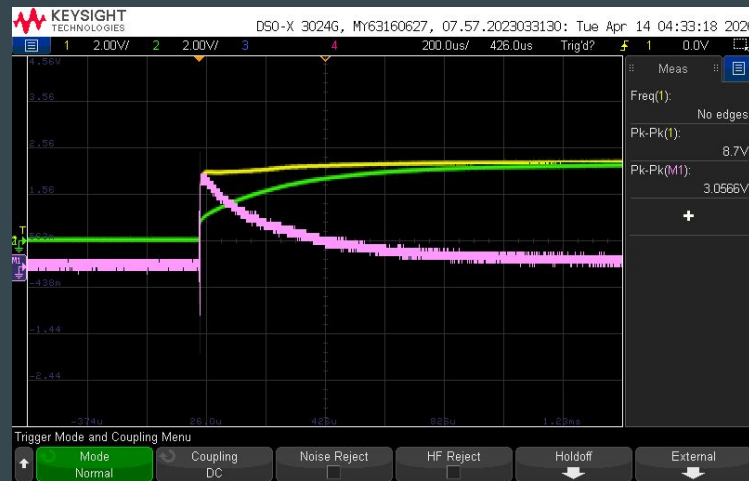


Fig. 6 100uF Decoupling

This measurement doesn't require averaging as it is a single transient signal. The peak current is 1.39A (using 100uF measurement). This is the current on the power rail to charge up the capacitance. This current has an exponential decay as it is short lived.

Summary

- A larger decoupling capacitor results in a longer startup
 - A greater inrush current is also a result of a larger decoupling capacitor given by $I=C*dv/dt$
 - Decoupling capacitors reduce noise but at a cost of power consumption and longer time constant.
 - Inrush current is unavoidable, so it is important to carefully design circuits to fit design needs.
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